SignBridge – Real-Time Sign Language Translator

This project aims to bridge the communication gap between the Deaf/Hard-of-Hearing community and the hearing population through a bi-directional, real-time AI translation interface.

Problem Statement

Deaf and hard-of-hearing individuals often face significant communication barriers in daily interactions (e.g., healthcare, retail, or emergency services) when a human sign language interpreter is unavailable. Existing solutions are often one-sided or rely on slow letter-by-letter fingerspelling. There is a critical need for a **single-interface, bi-directional tool** that translates dynamic sign language into text/speech and spoken words into visual signs or text instantly.

Functional Requirements

Sign-to-Text (Deaf User)

- **Real-time Gesture Capture:** Access the device camera to stream high-frame-rate video for motion analysis.
- **Landmark Tracking:** Utilize computer vision to track 21 hand points, facial expressions, and body pose (Holistic tracking).
- **Dynamic Word Recognition:** Recognize full-word gestures (sequences of motion) rather than just static letters.
- **Text/Speech Output:** Convert recognized gestures into text bubbles and provide an optional Text-to-Speech (TTS) feature for the hearing user.

Speech-to-Sign (Hearing User)

- **Voice Recognition:** Capture spoken audio via microphone and convert it to text using Speech-to-Text (STT) technology.
- **Visual Representation:** Display the translated response to the Deaf user through an animated 3D avatar or a pre-recorded video library of signs.
- **Grammar Translation:** (Optional) Use a language model to convert spoken English into "Sign Glosses" (the conceptual order used in sign language).

System & UI

- **Single Unified Interface:** Provide a split-view or overlay system where both users can see the camera feed and the chat history simultaneously.
- **Low Latency:** Ensure translation happens in under 2 seconds to maintain the flow of natural conversation.
- **Privacy-First Processing:** Perform landmark detection and gesture classification on-device (client-side) to ensure camera data is not stored on a server.

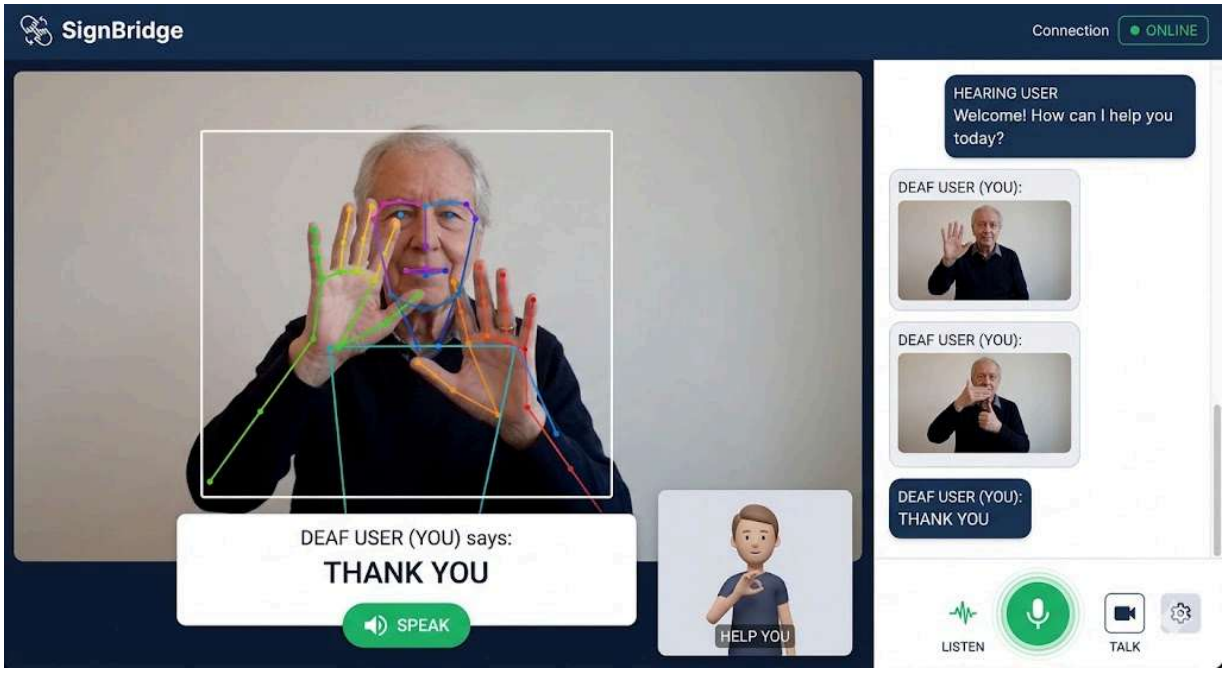
Implementation Roadmap (No Dataset Strategy)

Since you are a full-stack developer, you can follow this "Builder's Path" to create your own dataset and model:

1. **Data Acquisition:** Use a script to record your own "Sign Skeletons." Instead of saving video, save the X, Y, Z coordinates from **MediaPipe Holistic** as CSV/NumPy files.
2. **Model Training:** Build a **Long Short-Term Memory (LSTM)** network. This model will take a sequence of 30 frames of coordinates and predict the word being performed.
3. **Frontend Integration:** Build the UI in **React** or **Next.js**. Use **TensorFlow.js** to load your trained model so the "brain" of the app runs directly in the user's browser.
4. **Refinement:** Use a small LLM (like Gemini) to take the raw predicted words and "clean them up" into grammatically correct sentences for the hearing user.

Technical Stack

Component	**Technology**
Frontend	React.js / Tailwind CSS
Vision/Tracking	MediaPipe Holistic
ML Framework	TensorFlow / Keras (Python for training, JS for web)
Speech-to-Text	Web Speech API
Translation Logic	LSTM (Long Short-Term Memory) Neural Network


SignBridge
Connection
ONLINE
HEARING USER
Welcome! How can I help you today?
DEAF USER (YOU):
DEAF USER (YOU):
DEAF USER (YOU):
THANK YOU
DEAF USER (YOU) says:
THANK YOU
SPEAK
HELP YOU
LISTEN
TALK